Machine Learning-Based Algorithm for 1-D Wave Spectrum Retrieval From SAR Imagery as Passing Oceanic Eddy

Yuyi Hu, Weizeng Shao, *Member, IEEE*, Xingwei Jiang, and Maurizio Migliaccio, *Fellow, IEEE*

***Abstract*—The inversion of the 1-D wave spectrum from dual-polarized synthetic aperture radar (SAR) data is performed using machine learning methods, namely, random forest (RF), eXtreme Gradient Boosting (XGBoost), support vector regression (SVR), and convolutional neural networks (CNNs). This process incorporates an improved hydrodynamic modulation transfer function (MTF), calibrated with more than 8000 Sentinel-1 (S-1) Ground Range Detected (GRD) images from 2021 to 2024, to account for shear currents induced by oceanic eddies. This study primarily aims to identify the differences between algorithms for 1-D wave spectrum retrieval from S-1 SAR image as passing oceanic eddy and examine the characteristics of sea surface waves change observed by SAR passing through oceanic eddies. SAR retrievals are compared with significant wave heights (SWHs) from Haiyang-2 (HY-2) altimeter and wave spectrum from the Surface Wave Investigation and Monitoring (SWIM) and National Data Buoy Center (NDBC) buoys. The XGBoost model is established as superior for retrieving 1-D wave spectra and SWH, recording the lowest root mean squared error (RMSE) (0.33 m versus SWIM, 0.30 m versus HY-2, and 0.68 m versus Buoy) and highest correlation in validation. Applied to a North Atlantic case study, it effectively derives sea states from the S-1 image. This high-resolution analysis reveals that oceanic eddies significantly amplify wave energy, resulting in elevated SWH inside their cores. While model accuracy declines in high kinetic energy eddies, the SAR-derived retrievals maintain reliability across eddy-affected regions.**

***Index Terms*—1-D wave spectrum, Oceanic eddy, oceanic eddy, synthetic aperture radar (SAR).**

I. Introduction

SEA surface wave is a widely observed phenomenon [1] in the ocean and plays a crucial role in offshore dynamics. The measurements from ship and buoy are the most reliable resource for oceanography at a regular sea state [2]. The mixing process of ocean waves has a significant impact on the exchange of matter and energy in mesoscale ocean phenomena. On the other hand, mesoscale phenomena can change the structure of the ocean current, thereby affecting the propagation direction and velocity of wave. Traditional bulk parameters like significant wave height (SWH) and mean wave period (MWP) are insufficient to characterize eddy–wave interactions. In contrast, the 1-D wave spectrum provides an essential perspective for analyzing the underlying energy-frequency structure of this process [3]. The research on interaction between wave and mesoscale dynamics (i.e., eddy and front) requires synchronized on-scene observations at a large scale. Therefore, two techniques, i.e., remote sensing and numeric models, are popularly used for analyzing the wave characteristics at mesoscale and largescale, especially at extreme sea state, i.e., tropical cyclone (TC) [4] and polar region [5].

During the 105-day mission of SEASAT in 1978, several sensors [6], [7] were tested. It is confirmed that remote sensing is an advanced technology for Earth observation. In general, sea surface winds and waves are operationally monitored by spaceborne sensors, including scatterometers and microwave radiometers for wind, altimeters and the surface wave investigation and monitoring (SWIM) instrument for waves, and synthetic aperture radar (SAR) for multidynamic sea surface observations. Scatterometers, i.e., Advanced Scatterometer (ASCAT) onboard MetOp [8] and Haiyang-2 (HY-2) [9], have the ability to measure wind vector (i.e., wind speed and direction) over global seas, whereas wind speed is detectable by microwave radiometer [10]. However, strong wind from the microwave radiometer is reliable during a TC [11] due to the saturation of backscattering signal from the scatterometer. Altimeters, i.e., Topex/Poseidon [12] and Jason-2/3 [13], are designed for sea level and SWH monitoring. Since 2018, SWIM onboard Chinese–French Oceanography SATellite (CFOSAT) has provided wave spectrum even during a TC [14]. The above remotely sensed products have a swath spatial resolution of 25 km, which does not meet the requirement for oceanographic research at submesoscale and mesoscale levels. SAR offers the distinct advantage of providing synchronized measurements of sea surface dynamics at high spatial resolution [15].

Extensive research in recent decades has been devoted to advancing SAR wind and wave retrieval algorithms. According

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to previous studies [16], [17], the SAR backscattering signal is positively proportional to sea surface wind at the angle between the radar beam and the sea surface greater than 20°. Based on this principle, a geophysical model function (GMF) relating backscattering signal and wind vector is developed for wind retrieval from various SARs, i.e., TerraSAR-X [18], RADARSAT-2 [19], Sentinel-1 (S-1) [20], and Gaofen-3 (GF-3) [21]. The difficulty in the application of GMF is prior information on wind direction, and the solution is popular employing the prediction from an atmospheric numerical model. Current approaches for ocean wave retrieval from SAR imagery fall into two categories: theoretical schemes, based on wave mapping mechanisms for data assimilation in numerical models, and empirical models, designed for near-real-time (NRT) operational sea state services [22]. Tilt, hydrodynamic modulation, and velocity bunching are the SAR mapping mechanisms [23]. The fundamental principle of the theoretical scheme is to minimize the cost function representing the spectral mapping from ocean waves to SAR intensity. This is the basis of existing algorithms, i.e., Max Planck Institute (MPI) [24], partition rescaling and shift algorithm (PRSA) [25], and parameterized first-guess spectrum method (PFSM) [26], which relies on the first-guess wave spectrum produced by a parametric wave function or numeric wave mode. In the context of artificial intelligence (AI), the cost function can be replaced by deep learning [3], [27]. The difference in SAR backscattering at the polarimetric channel is determined by tilt, because hydrodynamic modulation and velocity bunching are independent on polarization [28]. Therefore, the wave slope spectrum is estimated from quad-polarized SAR measurement [29]. The advantage of empirical model avoids calculating complex modulation transfer functions (MTFs) of three modulations, which directly relates wave parameters with SAR image variables [30], [31], [32]. The field has evolved in recent studies [33], [34], [35] from empirical models to deep learning techniques for SAR wave parameters retrieval. Results demonstrate that both paradigms achieve comparable reliability, with SWH errors typically under 0.3 m.

Oceanic eddy is a bridge for energy exchange between oceanic dynamics at small and large scales [36]. As surface waves traverse mesoscale eddies, wave characteristics are significantly modulated by the effects of rotational currents and frontal gradients at the eddy boundary [37]. The modulation of surface waves by oceanic eddies arises from the complete horizontal velocity gradient, which includes the eddy's velocity magnitude/direction and shear components like divergence, vorticity, and strain [38]. Through the reconstruction of satellite altimetry-derived surface current fields, it is found that the swell propagation will be altered by the action of mesoscale eddies in the Southern Ocean [39], [40]. Additionally, numerical studies demonstrate that mesoscale oceanic eddies significantly modulate wave characteristics (height, period, and direction) through current-induced refraction and submesoscale instabilities while concurrently altering surface winds, creating coupled feedbacks that amplify wave variability across scales [41], [42]. The modulation effects of mesoscale eddies on sea surface wave fields in South China Sea are investigated using SWIM measurements [43].

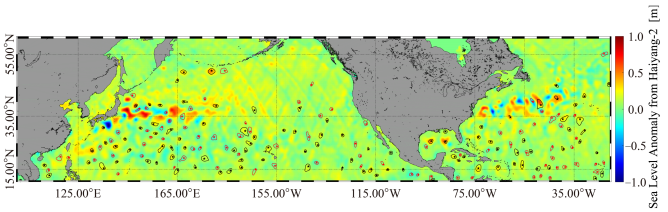

Fig. 1. Daily SLA from HY-2 altimeter constellation on July 21, 2024, in which the red and black circles represent the CEs and anti-CEs (AEs), respectively.

Collectively, the characteristic of sea surface wave is interesting issue as passing through oceanic eddy using remotely sensed data with high spatial resolution.

Although S-1A/1B Wave Mode products have been extensively employed in ocean wave studies [44], [45], their limited spatial resolution hinders the analysis of wave characteristics influenced by oceanic eddies. Therefore, the development of specialized algorithms for retrieving wave spectra from SAR imagery in eddy-rich environments is still needed. In this study, utilizing daily sea level anomaly (SLA) from HY-2 altimeter, oceanic eddies are identified. Based on identification results, S-1 images covering oceanic eddies are collected. Then, machine learning is implemented for 1-D wave spectrum from S-1 image using the collocation with simulation from WAVEWATCH-III (WW3), in which the MTF of hydrodynamic modulation considering shear current generated by oceanic eddy is improved. SAR retrievals are validated against measurements from HY-2 altimeter, National Data Buoy Center (NDBC) buoys, and SWIM. Finally, utilizing SAR-derived wave spectrum, the characteristic of sea surface wave is analyzed as passing through an oceanic eddy. The remainder of this article is organized as follows. Section II introduces the dataset. The methodology for oceanic eddy identification and schemes of machine learning methods is given in Section III. Section IV gives the validation of SAR-derived wave spectrum and the characteristics of sea surface wave as passing through oceanic eddy is studied in Section V. Finally, the conclusion is placed in Section VI.

II. Description of Datasets

In this study, the dataset is grouped into three categories, i.e., daily SLA data from HY-2 with dual-polarized S-1 image, hindcasting wave by WW3, and validation sources comprising from HY-2 altimeter, NDBC buoys, and SWIM.

A. Daily SLA Data and S-1 Image

Since 2018, Chinese civilian satellites, called HY-2B, which carries scatterometer and altimeter, have started to distribute on-orbit products. HY-2 C and HY-2D that are sisters of HY-2B are continuously operating since 2021. Moreover, daily SLA data derived from HY-2 constellation are generated by the National Satellite Ocean Application Service (NSOAS) from 2021 to present. In our previous study [46], characteristics of oceanic eddy identified through HY-2 and Aviso daily SLA data by vector geometry algorithm is analyzed. Mesoscale eddies with a radius greater than 25 km were identified from

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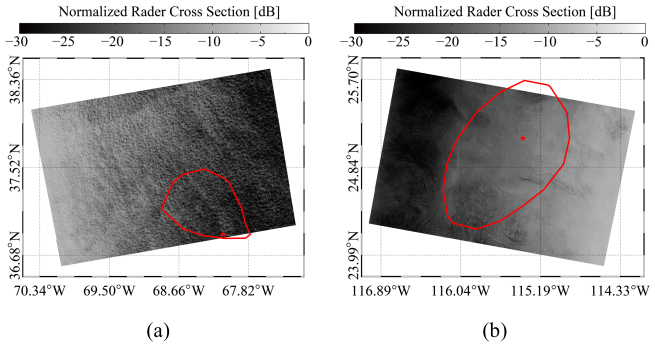

Fig. 2. Two cases of VV polarization Level-1 GRD S-1 SAR image acquired in IW mode on (a) November 9, 2024 at 22:34 UTC and (b) January 29, 2024 at 13:39 UTC. The red circle represents the eddy boundary edge line identified by HY-2 SLA.

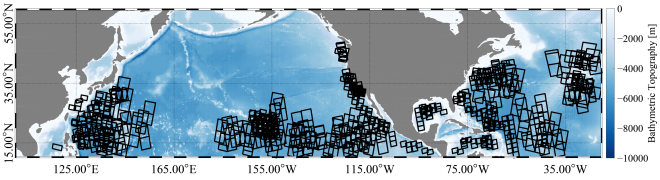

Fig. 3. Geographic locations of collected S-1 images that covered oceanic eddies.

HY-2 SLA data for the period 2021–2024. Fig. 1 presents the daily SLA map from HY-2 on July 21, 2024, where cyclonic eddies (CEs) and anticyclonic eddies (AEs) are denoted by red and black circles, respectively. Based on the detection results, >8000 dual-polarized [vertical–vertical (VV) and vertical–horizontal (VH)] S-1 Level-1 Ground Range Detected (GRD) images covering these eddies were collected. The selected images, acquired in interferometric wide (IW) and extra wide (EW) swath modes, were subsequently processed through radiometric calibration and thermal noise removal using (1) from the S-1 User Handbook [3].

The pixel size of IW image is 10 m. The swath coverage of IW image is 250 km, while the swath coverage of EW image can be up to 400 km. In this study, it should be noted that partial images have been used to study the influence of oceanic eddy on SAR wind retrieval [47]

$$\sigma_0 = \frac{DN^2 - \eta}{A^2} \tag{1}$$

where σ_0 is the normalized radar cross section (NRCS), η is the noise vector, and A is the calibration vector. As a case, Fig. 2 shows two VV polarization images acquired in IW mode on November 9, 2024 at 22:34 UTC [Fig. 2(a)] and January 29, 2024 at 13:39 UTC [Fig. 2(b)], where the red circle represents the eddy boundary edge line identified by HY-2 SLA. Geographic locations of collected S-1 images are illustrated in Fig. 3. It is worth noting that all collected S-1 images are from the North Pacific and North Atlantic. This selection was made because the vector geometry-based eddy detection algorithm is unsuitable for low latitudes (0°–5°), and these regions feature strong western boundary currents while providing reliable wave spectrum observations from NDBC buoys for validation.

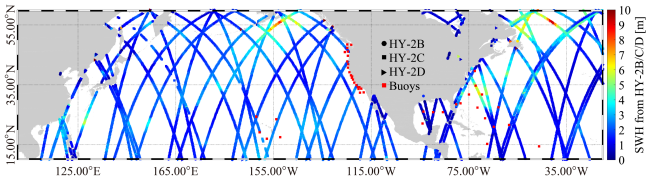

Fig. 4. SWH measured by HY-2 constellation on September 1, 2023, in which the red square represents the location of the NDBC Buoys.

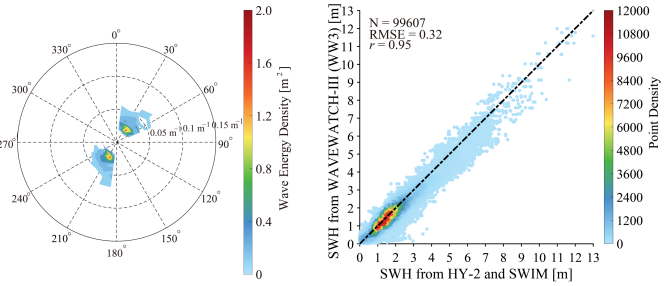

Fig. 5. (a) 2-D wave spectrum measured by the SWIM, which is located at [134.31°E, 13.81°N] on July 10, 2023. (b) Comparison between the WW3 simulation and the measurement of HY-2 and SWIM during 2024.

B. Hindcasting Wave

Machine learning requires an ambulant training dataset. Currently, moored buoys and SWIM can measure wave spectrum; however, few spectral observations are collocated with satellite imagery. Therefore, a third-generation numeric wave model (WW3) is employed to hindcasting waves, which is publicly released by the National Oceanic and Atmospheric Administration (NOAA). The forcing field is European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis (ERA-5) wind at 0.25 grids and intervals of 1 h. In addition, open boundary conditions are daily Copernicus Marine Environment Monitoring Service (CMEMS) sea surface current and sea level at grids of 0.08. The water depth is extracted from the General Bathymetry Chart of the Oceans (GBCO). The parameterizations of WW3 are described in our previous study [48], and these are not repeated here. The output of WW3 at 0.25 grids and intervals of 0.5 h is 1-D wave spectrum, in which the frequency ranges from 0.04118 to 0.7186 Hz. In this study, the time range of the WW3 model simulation is from 2021 to 2024, and the area is the coverage range of the collected S-1 images in Fig. 3.

C. Validation Sources

The altimeter can measure SWH following the footprints of flight (~10 km), i.e., Jason-3 and HY-2. The latitude of Jason-3 is up to 65°N, while the latitude of HY-2 reaches 85°N. These products are valuable for the research on oceanography, especially under extreme sea state such as Arctic Ocean [5], [49]. The aim of SWIM is to obtain a wave spectrum with a spatial resolution of 18 km over global seas. In order to validate the SAR retrievals, the on-orbit measurements from HY-2 altimeter, NDBC buoys, and SWIM within the swath coverage of SAR images are collected. Fig. 4 shows the SWH map from the HY-2 altimeter on September 1, 2023, where

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the red square represents the location of the NDBC Buoys. Fig. 5(a) shows the 2-D wave spectrum measured by SWIM, which is located at [134.31°E, 13.81°N] on July 10, 2023. It should be noted that 180° ambiguity of wave propagation exists in the SWIM product; therefore, the 1-D wave spectrum is half of integrating the 2-D whole wave spectrum in the omnidirectional. The validation of WW3 simulation with the measurement of HY-2 and SWIM was conducted in Fig. 5(b). It should be noted that we matched nearly 100 000 matching points during 2024. The comparison result yields a 0.32-m root-mean-squared error (RMSE) and a 0.95 correlation (r), which reveals the simulation results are accurate and suitable for this study.

III. Methodology

In this section, the improvement of hydrodynamic MTF is presented. Then, the schemes of four traditional machine learning methods, i.e., random forest (RF), eXtreme Gradient Boosting (XGBoost), support vector regression (SVR), and convolutional neural networks (CNNs), are described. Finally, the training process of machine learning method is exhibited.

A. Improvement of Hydrodynamic MTF

The SAR imaging of the ocean surface is a complex function of multiple factors. The dominant backscatter mechanism shifts from Bragg scattering under moderate winds to non-Bragg processes (e.g., wave breaking and specular reflection) in storms, with additional complexities in coastal zones. The imaging is further nonlinearly distorted by the SAR transfer function, primarily velocity bunching. Furthermore, sea surface currents modulate wave properties through wave–current interaction, thereby altering the surface roughness that SAR ultimately detects. Under the background of strong shear current caused by oceanic eddy, hydrodynamic modulation supposedly plays a more important role. The existing MTFs of tilt modulation T^t [see (2)], velocity bunching T^{vb} [see (3a)], and hydrodynamic modulation T^{hy} [see (4)] are all based on theoretical derivation

$$T^t = ik_l \frac{4\cot\theta}{1 \pm \sin^2\theta} \tag{2}$$

$$T^{vb} = k_l \frac{R}{V} T^v \tag{3a}$$

$$T^v(\omega, \theta, \phi) = \omega(\sin\theta\cos\phi + i\cos\theta) \tag{3b}$$

$$T^{hy} = 4.5\ k\omega \frac{0.5\ i - \omega}{\omega^2 + 0.25} \sin^2\phi \tag{4}$$

where k is the wavenumber, k_l is the wavenumber in the radar direction, ω is the angular frequency of the sea surface microscale wave, θ is the radar incidence angle, φ is the wave direction relative to the azimuth direction, R is the apparent slant range of the radar, and V is the flight direction of the satellite platform. The mapping relationship between the wave spectrum $S(\omega)$ and the SAR intensity spectrum $E(\omega)$ is as follows:

$$E(\omega) = S(\omega) \times \left(T^t + T^{vb}_{\omega} + T^{hy}\right). \tag{5}$$

TABLE I
Values of Constants p and q in (6)

Constant	Value	Constant	Value
p_0	51.763	q_0	1.030
p_1	−149.655	q_1	−8.303
p_2	−4.273	q_2	−0.957
p_3	−2.629	q_3	−1.546

The theoretical derivation of hydrodynamic MTF deviates from reality in the context of oceanic eddy. Here, the formulation of hydrodynamic MTF is modified as follows:

$$T^{hy} = p_0 + p_1\omega + p_2 k + p_3 \sin\phi + (q_0 + q_1\omega + q_2 k + q_3 \sin\phi)\, i \tag{6}$$

where p and q are constants, as listed in Table I. Specifically, a subimage with 256×256 pixels is first extracted from the original SAR data to obtain its NRCS map. This NRCS matrix is then subjected to a 2-D fast Fourier transform (2D-FFT) to generate the VV-polarized SAR intensity spectrum. Based on the VV-polarized SAR intensity spectrum, the simulation takes the WW3-simulated spectrum and (2), (3) and (5) into (6), and p and q are conveniently tuned by regression fitting.

B. Random Forest

As a type of supervised learning algorithm, RF is an ensemble learning method consisting of a classifier or regressor composed of multiple decision trees. First, RF uses the bootstrap resampling technique to repeatedly randomly sample from the original set with replacement in order to produce a new training set. Then, based on these self-service sets, multiple decision trees are generated so as to reduce the risk of overfitting the model. As for the classification problem, the final prediction result is determined by majority voting. In regression problems, the model's output is determined by averaging predictions from all constituent trees. The steps of RF are given as follows.

1) Generate M subsets D_m from the original set $D = (X_I, Y_I)\}_{I=1}^{N}$ with replacement

$$D_m \leftarrow \text{Bootstrap}(D), |D_m| = N. \tag{7}$$

1) For each subset D_m, build a regression tree $T_m(x))$ and minimize the mean squared error for split (j, s)

$$\min_{j,s}\left[\sum_{x_i \in R_L}(y_i - \bar{y}_L)^2 + \sum_{x_i \in R_R}(y_i - \bar{y}_R)^2\right] \tag{8}$$

where R_L and R_R represent the left and right regions, respectively, and $\bar{y}_L$ and $\bar{y}_R$ denote the mean values of y in those regions.

1) Predict $\hat{F}(x)$ as the average of M trees

$$\hat{f}(x) = \frac{1}{M}\sum_{m=1}^{M} T_m(x) \tag{9}$$

$$T_m(x) = \sum_{j=1}^{J_m} c_{jm} I\left(x \in R_{jm}\right) \tag{10}$$

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where $I(x \in R_{jm})$ is an indicator function for whether x falls in the region R_{jm}.

1) Evaluate the performance of the aggregated model using appropriate metrics like RMSE

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}. \tag{11}$$

C. EXtreme Gradient Boosting

XGBoost is a powerful algorithm that aims to enhance model prediction performance through iterative optimization by constructing multiple weak learners, typically decision trees. The XGBoost efficiently manages model complexity while boosting prediction accuracy by incorporating regularization terms and optimizing parallel computation. The model utilizes a forward stepwise approach, generating decision trees by minimizing a loss function that includes regularization terms. Each new tree is designed to correct the residuals of the current model, aligning with the negative gradient direction of the loss function. During node splitting, the algorithm utilizes either an exact greedy approach or an approximation method to identify optimal split points, maximizing information gain for feature selection. Additionally, XGBoost includes a built-in function to evaluate feature importance, which measures each feature's contribution based on the cumulative gain during node splitting. The following are the steps involved in XGBoost.

1) At iteration t, the model is an additive combination of trees

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i) \tag{12}$$

where $f_t(x_i)$ is the prediction of the tth tree.

1) The regularized objective function includes loss and regularization terms

$$L^{(t)} = \sum\nolimits_{i=1}^{N} L\left(y_i, \hat{y}_i^{(t)}\right) + \sum\nolimits_{k=1}^{t} \Omega(f_k) \tag{13}$$

$$\Omega(f_k) = \gamma T_k + \frac{1}{2}\lambda \parallel w_k \parallel^2 \tag{14}$$

where T_k is the number of leaves and w_k is the leaf weight.

1) Approximate the loss function using the Taylor expansion

$$L^{(t)} \approx \sum_{i=1}^{N}\left[g_i f_t(x_i) + \frac{1}{2}h_i f_t^2(x_i)\right] + \Omega(f_t) \tag{15}$$

$$g_i = \partial_{\hat{y}_i^{(t-1)}} L\left(y_i, \hat{y}_i^{(t-1)}\right) \tag{16}$$

$$h_i = \partial^2_{\hat{y}_i^{(t-1)}} L\left(y_i, \hat{y}_i^{(t-1)}\right). \tag{17}$$

4) The final model is the sum of all trees

$$\hat{y}_i = \sum_{t=1}^{T} f_t(x_i). \tag{18}$$

D. Support Vector Regression

SVR is an adaptation of support vector machines (SVMs) for regression tasks, serving as a canonical model for solving regression problems. The algorithm maps samples from the original feature space into a higher dimensional space via kernel functions, fitting a robust regression hyperplane within this transformed space. The hyperplane (w, b) is defined as follows:

$$w^T x + b = 0 \tag{19}$$

where w is the weight vector determining the hyperplane's orientation and b is the bias term controlling its offset relative to the origin. The regression hyperplane is thus fully determined by w and b.

Unlike traditional regression models that compute loss directly from the difference between model output $f(\mathrm{x})$ and true value y, SVR permits absolute errors up to ε. This establishes a width-$2\varepsilon\varepsilon$ - insensitive tube where predictions within $\mid f(\mathrm{x}) - y \mid \leq \varepsilon$ incur zero loss, while those beyond $\mid f(x) - y \mid > \varepsilon$ incur linear loss. Through the introduction of slack variables and Lagrange multipliers, the SVR solution takes the form

$$f(x) = w^T \phi(x) + b = \sum_{i=1}^{m}(\hat{\alpha}_i - \alpha_i)\kappa(x, x_i) + b \tag{20}$$

where $\kappa(x_i, x_j) = \phi(x_i)^{\mathrm{T}}\phi(x_j)$ is the kernel function and $\hat{\alpha}_i$ and α_i are Lagrange multipliers. The kernel trick circumvents the curse of dimensionality while reducing computational complexity.

E. Convolutional Neural Networks

CNNs are deep feedforward neural networks characterized by local connectivity and weight sharing. Particularly effective for image processing tasks, they rank among the most extensively deployed models today. A standard CNN architecture integrates: convolutional layers, pooling layers, fully connected layers, activation functions (most commonly ReLU, tanh, and sigmoid), and output layers. The key structural characteristics encompass: 1) local connectivity; 2) weight sharing; and 3) pooling operations. The 1-D convolution operation between data matrix x and w kernel is defined as

$$y = x * w \tag{21}$$

where $*$ denotes the convolution operator. Within convolutional layers, the pre-activation output z_l is computed as follows:

$$z_l = w_l * a_{l-1} + b_l \tag{22}$$

where w_l is the learnable kernel, b_l is the learnable bias vector, and a_{l-1} represents the activations from layer l–1.

In pooling layers, dimensionality reduction is performed to compress data and reduce parameters, mitigating overfitting through spatial downsampling. Fully connected layers subsequently integrate all local features into global representations, evaluating each feature's contribution to the final output.

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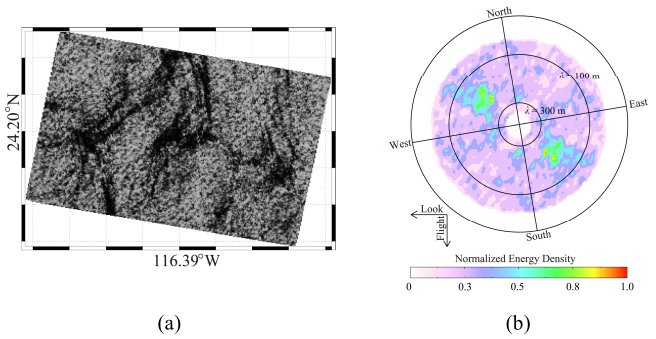

Fig. 6. (a) Subscene images were extracted from the VV-polarized SAR images shown in Fig. 2(b). (b) Corresponding SAR intensity spectrum.

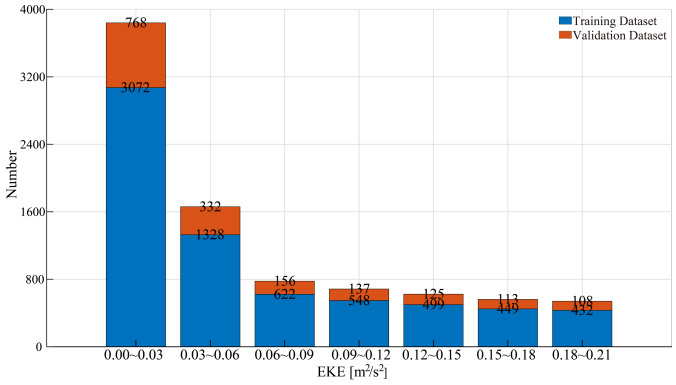

Fig. 7. Distributions of S-1 image numbers versus the EKE in different datasets.

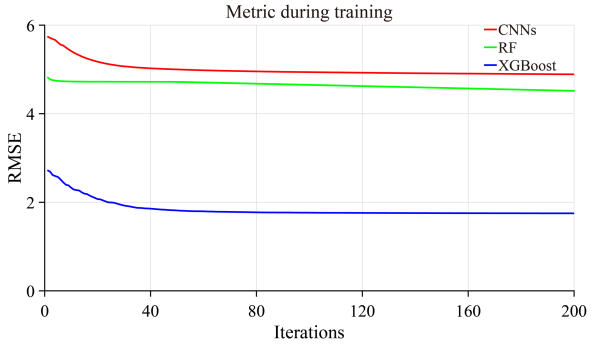

Fig. 8. Performance of three machine learning methods training process.

IV. Results

In this section, the model training process and the validation of the SAR retrieval wave spectrum are investigated. Subsequently, the error analysis of the SAR wave retrievals by the herein method is conducted.

A. Training Process

Before training the model, the data were filtered to remove outliers. During prediction, the input parameter was identified as outliers and set such data points to invalid values. The inputs include VV-polarized SAR intensity spectrum and three MTFs. The simulations from WW3 are treated as ground truth. S-1 images were divided into 256 × 256 pixels, i.e., 2.6 × 3 km for IW image and 6.4 × 6.4 km for EW image

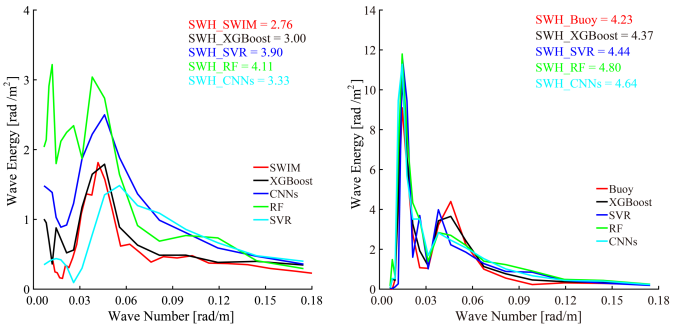

Fig. 9. (a) Wave spectrum derived from SAR observations by four models at 17:33 UTC and measured by SWIM at 18:07 UTC on March 6, 2023. (b) Wave spectrum derived from SAR observations by four models at 14:01 UTC and measured by NDBC buoy at 14:00 UTC on June 10, 2021.

in range × wide directions. As an example, the process of the SAR intensity spectrum form was shown in Fig. 6. It should be noted that the range of recognizable wavelengths is 20 ~ 1000 m. Fig. 6(a) presents the quick-look of a subscene in Cartesian coordinate extracted from the image in Fig. 2(b), and Fig. 6(b) reveals the matching SAR intensity spectrum in polar coordinate. Fig. 7 presents the distributions of S-1 image numbers versus the eddy kinetic energy (EKE) in the validation and training datasets. The results reveal that the machine learning-based algorithms are trained by 6500 images and the other over 2000 S-1 images are used for validation. The SAR subscenes located at WW3 grid points are grouped, with each group containing cases where the time difference between the WW3 simulations and SAR imaging is less than 15 min. Overall, more than 200 000 samples were collected from WW3 simulations to form the training dataset. Fig. 8 presents the performance of the above proposed model training process, i.e., XGBoost, CNNs, and RF. It should be noted that the SVR lacks assessment of training process. The results show that all three machine learning models eventually achieve convergence during training, with XGBoost achieving RMSE values below 2.1, while CNNs and RF reach an RMSE of 4.7. Overall, the evaluation confirmed the notable efficacy of the XGBoost model.

B. Validation of the SAR-Derived Wave Spectrum

The four well-trained typical models described above were implemented to invert the 1-D wave spectrum from over S-1 images in eddy-affected areas. Fig. 9(a) presents an illustrative example of an SAR-derived wave spectrum by four models collocated with the SWIM product. It is noted that the 2-D SWIM wave spectrum was directionally integrated and then interpolated to the wavenumber grid of the collocated SAR-derived wave spectrum. The comparison results reveal that the 1-D wave spectrum by the XGBoost method is most consistent with SWIM observation results. In addition, the SWH retrieval by XGBoost is closest to SWIM observation results with a bias of 0.24 m. Similarly, an example of an SAR-derived wave spectrum by four models collocated with the buoy measurement was conducted in Fig. 9(b). The comparison results reveal that all methods exhibit a similar

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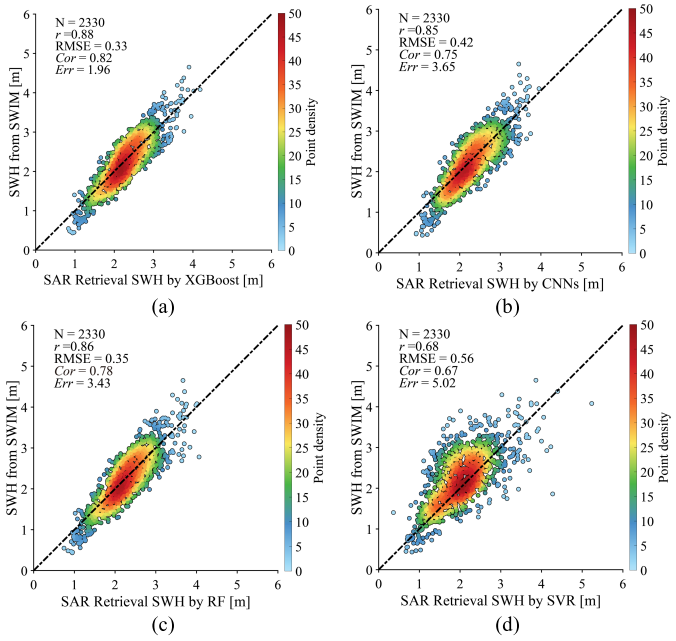

Fig. 10. Comparison between SAR-retrieved results obtained using (a) XGBoost, (b) CNNs, (c) RF, and (d) SVR against collocated SWIM observations.

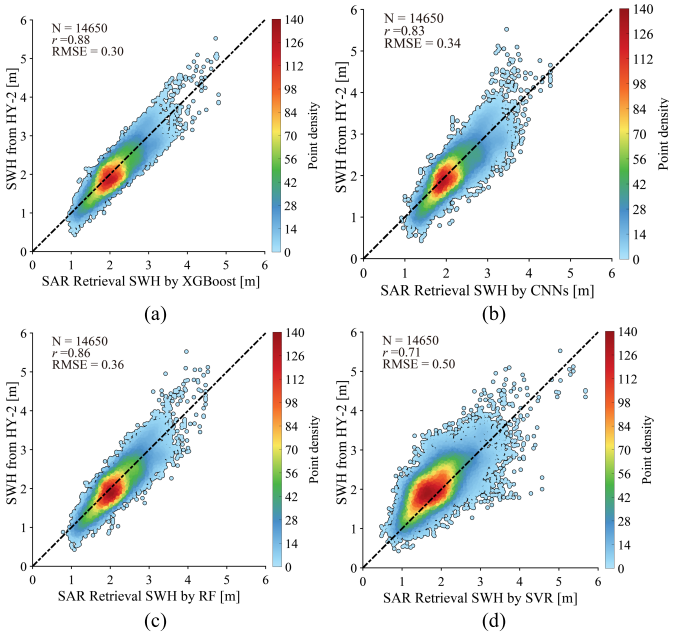

Fig. 11. Comparison between SAR-derived results obtained using (a) XGBoost, (b) CNNs, (c) RF, and (d) SVR against collocated HY-2 measurements.

wave energy distribution pattern characterized by two complex wave systems. Furthermore, the XGBoost exhibits higher correlation in the high wavenumber region than the other three machine learning methods (RF, SVR, and CNNs). The bias between the SWH retrieval by XGBoost and NDBC buoy measurement is 0.14 m, which is better than the other three machine learning methods (RF, SVR, and CNNs) with the bias >0.2 m. Overall, the XGBoost-retrieved 1-D spectrum shows the best consistency with the SWIM observations in terms of spectral shape.

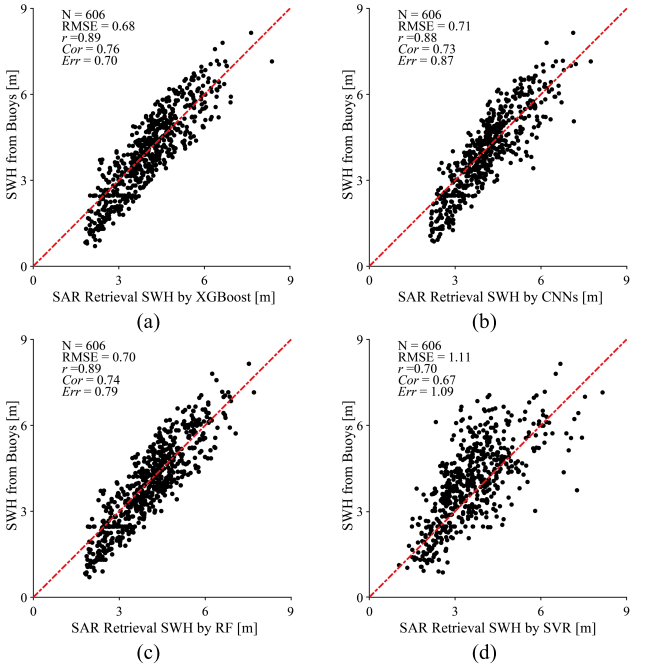

Fig. 12. Comparison between SAR-retrieved results obtained using (a) XGBoost, (b) CNNs, (c) RF, and (d) SVR against collocated measurements from NDBC buoys.

Statistical analysis of SAR retrievals from over 2000 S-1 images showed that XGBoost achieved the lowest RMSE (0.33 m) for SAR-derived SWH compared to SWIM measurements, with 0.88 r [Fig. 10(a)]. In comparison, the CNN model [Fig. 10(b)] achieves an RMSE of 0.42 m, with 0.85 r, the RF model [Fig. 10(c)] yields an RMSE of 0.35 m, with 0.86 r, and the SVR model [Fig. 10(d)] acquired an RMSE of 0.55 m, with 0.78 r. Additionally, the XGBoost model demonstrated reliable wave spectrum retrieval when validated against SWIM measurements, achieving superior performance (Cor = 0.82 and Err = 1.96). This result is better than the other three models (CNNs with 0.75 Cor and 3.65 Err, RF with 0.78 Cor and 3.43 Err, and SVR with 0.67 Cor and 5.02 Err). Similarly, Fig. 11(a) validates the SAR-derived SWH against HY-2 measurements, with XGBoost achieving an RMSE of 0.30 m and r of 0.88. This demonstrates superior performance compared to the alternative three methods, i.e., an RMSE of 0.54 m and 0.89 r by CNNs [Fig. 11(b)], an RMSE of 0.34 m and 0.83 r by RF [Fig. 11(c)], and an RMSE of 0.50 m and 0.71 r by SVR [Fig. 11(d)]. The SAR-derived SWH was benchmarked against NDBC buoy measurements (Fig. 12). The comparison demonstrates that the XGBoost model achieved the best performance (r = 0.89 and RMSE = 0.68 m), surpassing the CNN (r = 0.88 and RMSE = 0.71 m), RF (r = 0.89 and RMSE = 0.70 m), and SVR (r = 0.70 and RMSE = 1.11 m) models. The lower error observed in HY-2 altimeter validation compared to NDBC buoy validation is to be expected, as SAR and altimeter data share common error sources such as rain cells and wave breaking. Based on the statistical analysis, the XGBoost model demonstrates superior performance for 1-D wave spectrum inversion among the tested methods. However, the herein algorithm has a slightly

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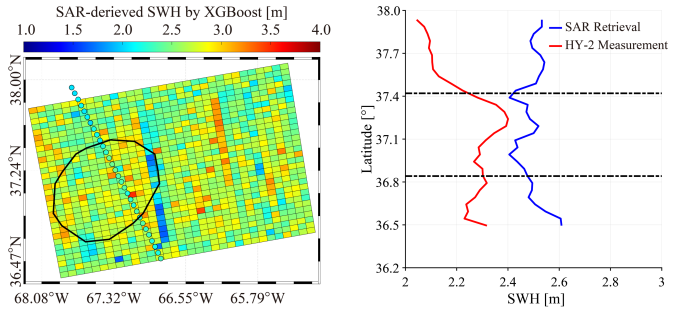

Fig. 13. (a) SAR-derived result map by XGBoost at 02:18 UTC on June 22, 2024 overlaid by the HY-2 measurement, in which the black line represents the eddy edge. (b) Trend in SWH along the latitude corresponding to the HY-2 measurement with the latitude following the HY-2 footprints.

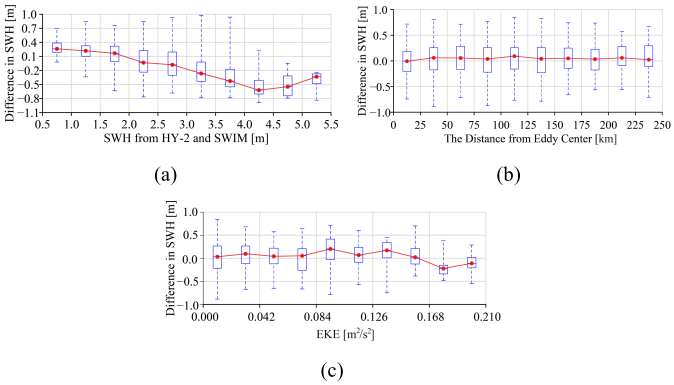

Fig. 14. Variation of difference (SAR retrievals minus matchups from HY-2 and SWIM) for SWH from HY-2 altimeters and SWIM at three parts. (a) Inside of eddy. (b) Edge of eddy. (c) Outside of eddy.

worse performance than a <0.3-m RMSE of SWH achieved by a one-step direct derivation framework (i.e., empirical models and deep learning methods).

C. Error Analysis

A case study in the North Atlantic Ocean illustrated the benefits of sea state observations by S1 based on the proposed model. Fig. 13(a) reveals an example of the SWH map by the XGBoost method at 02:18 UTC on June 22, 2024, overlaid with the HY-2 measurement. It was found that the variation in SWH by SAR retrieval is consistent with the HY-2 measurement [Fig. 13(b)]. Furthermore, the SWH trend along the latitude reveals that the SAR retrieval in the inside of eddy has the better performance than the result at the outside of eddy. In addition, boxplots were used to evaluate the error distribution of wave retrievals from the XGBoost method. Fig. 14 shows the variation in differences between SAR-derived SWH and altimeter measurements (HY-2 and SWIM), along with their relationships to distance from eddy center and EKE. The comparison in Fig. 14(a) reveals that the overall difference in SWH shows a slight upward trend as the SWH increases (SWH <4 m), while the difference in SWH is decreasing at SWH >4 m. The accuracy of NRT SAR-derived SWH is approximately 0.25 m in open ocean and swell-dominated regions, as established in previous studies [33], [34], [50].

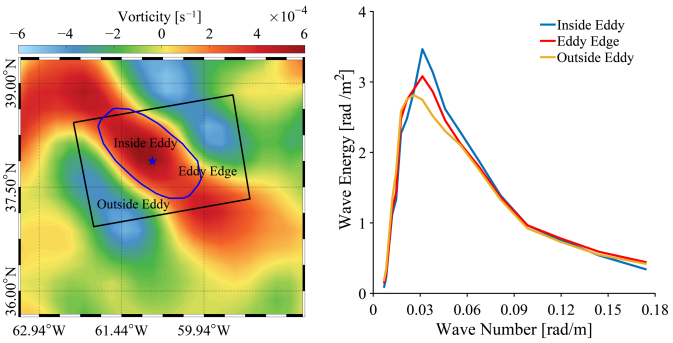

Fig. 15. (a) Illustration of the definition of different eddy locations covering surface current vorticity on June 10, 2024. (b) Comparison of mean 1-D wave spectrum energy at different eddy locations (blue: inside, red: edge, and orange: outside).

TABLE II
VARIATIONS IN MEAN SWH AT DIFFERENT EDDY LOCATIONS IN DIFFERENT SENSORS

Sensors		Difference of SWH (m)	Percentage variation of SWH
SAR	Inside-outside	0.21	8.12%
	Inside-edge	0.07	2.71%
	Edge-outside	0.15	5.80%
HY-2	Inside-outside	0.37	13.52%
	Inside-edge	0.28	10.32%
	Edge-outside	0.09	3.20%
SWIM	Inside-outside	0.45	19.75%
	Inside-edge	0.09	3.88%
	Edge-outside	0.36	15.87%

The greater deviation observed in the present algorithm is postulated to result from wave–current interaction effects. Fig. 14(b) presents the relationship between the difference and the distance from the eddy center. The result reveals that most of the data points are concentrated between a difference value of –0.5 and 0.5 m, and the algorithm has good performance in different eddy influence ranges. As shown in Fig. 14(c), the difference in SWH remains unchanged as EKE increases (EKE <0.084 m^2/s^2). However, the difference in SWH shows a huge change under the high EKE range. This indicates that the accuracy of 1-D wave spectrum by the herein algorithm will be affected by the strong eddy effect.

V. Discussion

SAR retrievals are employed to analyze the characteristics of sea surface wave as they pass through an oceanic eddy. As shown in Fig. 15(a), three characteristic positions relative to the eddy were defined [51]: the interior (area of homogeneous vorticity within the blue border), the edge (area of changing vorticity at the blue border), and the exterior (the remaining image area excluding other eddies). Fig. 15(b) presents the mean SAR-derived 1-D wave spectrum at these positions.

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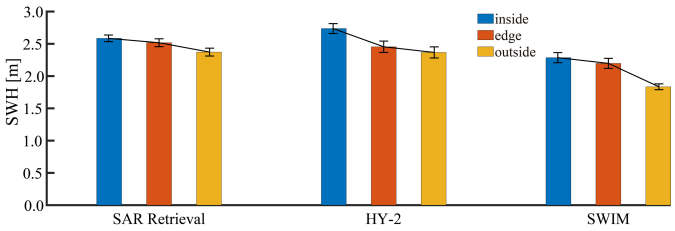

Fig. 16. Comparison of mean SWH at different eddy locations (blue: inside, red: edge, and orange: outside) from different sensors (SAR retrieval, HY-2, and SWIM).

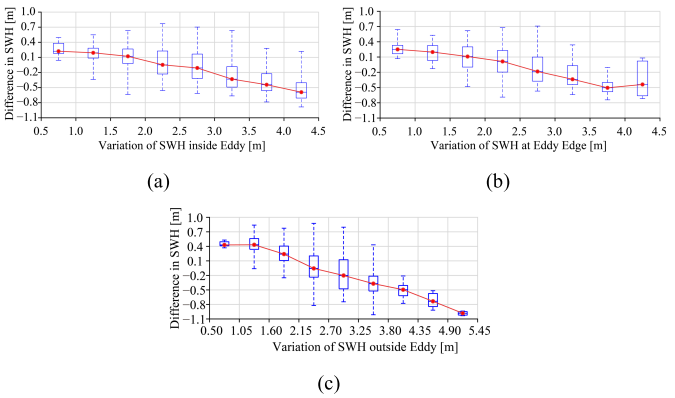

Fig. 17. Variation of difference (SAR retrievals minus matchups from HY-2 and SWIM) for SWH from HY-2 altimeters and SWIM at three parts. (a) Inside of eddy. (b) Edge of eddy. (c) Outside of eddy.

The comparison reveals that the wavenumber of maximum energy variation falls primarily within the 0.03–0.07-rad/m range under eddy influence. Moreover, eddies significantly enhance ocean wave energy. To further analyze the wave energy change at three eddy characteristic positions, the mean SWH from various sensors (SAR retrieval, HY-2, and SWIM) was calculated in Fig. 15. Comparison results (Fig. 16) reveal that the mean SWH decreases from the inside to the outside of eddy, which is similar to the result in [39]. In addition, Table II lists the mean variation in SWH at three eddy characteristic positions. The maximum variation in SWH occurred between the inside and outside of eddy. Collectively, it is concluded that SAR retrieval with high spatial resolution is useful to analyze the characteristics of waves passing through eddies.

To further assess the impact of eddies on the performance of the proposed algorithm, variations in SWH at three characteristic eddy positions were investigated, as shown in Fig. 17. The comparison results reveal that the variations in SWH are mostly concentrated between –0.5 and 0.5 m, and the proposed algorithm has a good applicability at three eddy characteristic positions. Additionally, it should be noted that the difference at the outside of eddy leads to the variation in the differences in SWH in Fig. 14(a). Overall, the variation in SWH shows a slight downward trend with the increase of SWH changes at three eddy characteristic positions.

VI. Conclusion

Oceanic eddies act as crucial bridges for energy exchange across scales, significantly modulating surface wave characteristics (SWH, wave period, and wave direction) as they traverse. This modulation is driven by eddy-induced rotational currents, frontal gradients, vertical velocity components, and velocity shear fields (encompassing divergence, vorticity, deformation, and strain), which dynamically alter wave propagation and energy. This study analyzed over 8000 S-1 images to investigate sea surface waves as they passed through oceanic eddies. Although the wave parameters (i.e., SWH and MWP) from our algorithm are less accurate than those from existing empirical or deep learning approaches, the model itself constitutes a significant advance in solving the persistent challenge of retrieving 1-D wave spectra from SAR. The conclusion is generally summarized as follows.

The hydrodynamic MTF, associated with shear currents generated by oceanic eddies, was improved using more than 6000 S-1 SAR images acquired in 2021–2024. The wave spectrum is retrieved from dual-polarization SAR images using machine learning methods (RF, XGBoost, SVR, and CNNs). These algorithms are trained on more than 6000 SAR images collocated with WW3-simulated wave spectrum. The SAR-derived retrievals are validated against SWH measurements from the HY-2 altimeter and wave spectrum from SWIM and NDBC buoys. XGBoost best retrieval 1-D wave spectrum from SAR, showing the highest consistency with SWIM and NDBC buoy data, with minimal SWH biases (0.24 and 0.14 m). Based on comprehensive statistical analysis against SWIM, HY-2 altimeter, and NDBC buoy measurements from over 2000 S-1 images, the XGBoost model demonstrates superior performance for 1-D wave spectrum and SWH retrieval. It consistently achieved the lowest RMSE (e.g., 0.33 m versus SWIM, 0.30 m versus HY-2, and 0.68 m versus Buoy) and the highest correlation coefficients (e.g., $r = 0.88$ and Cor = 0.82) compared to the CNN, RF, and SVR models, confirming its robustness and accuracy for SAR-derived ocean wave inversion.

1) A North Atlantic case study confirms the effectiveness of the XGBoost model for retrieving sea state from S-1 data. The SAR-derived SWH map aligns well with HY-2 altimeter measurements. Performance is better inside oceanic eddies than outside. Error analysis shows most SWH differences with HY-2 altimeters and SWIM are within ± 0.5 m, and performance is stable across different distances from the eddy center. However, the accuracy of the XGBoost algorithm is affected under conditions of high EKE (EKE >0.084 m^2/s^2), indicating a limitation in strongly dynamic eddy fields.
2) Based on the analysis of SAR retrievals, the characteristics of sea surface waves as they pass through oceanic eddies, defined by three characteristic positions: inside, edge, and outside, was investigated. The results demonstrate that eddies significantly increase ocean wave energy, with spectral energy variation concentrated in the 0.03–0.07-rad/m wavenumber range. Furthermore, the mean SWH decreases from the inside to the outside of an eddy. The proposed retrieval algorithm shows good applicability across all three eddy positions, with most SWH variations concentrated within a ± 0.5-m range. This confirms the utility of high-resolution SAR data for analyzing wave–eddy interactions. The overall SWH

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variation shows a slight downward trend with increasing SWH across the different eddy zones.

Although the proposed algorithm demonstrates good performance for wave spectrum inversion in eddy regions, the WW3-simulated wave spectrum still lacks sufficient accuracy for representing true wave conditions. SAR is well-established for measuring complex sea states, including wind waves and swell. This study preliminarily confirms that 1-D wave spectra can be inverted from SAR imagery using a physics-informed AI technique. Future improvements to this machine learning algorithm will incorporate data from NDBC moored buoys. A key direction for future research is to determine whether SAR's high spatial resolution can be leveraged to characterize regional wind wave and swell climate. In particular, the SAR-derived wave spectrum is expected to be used in developing an AI-based model, providing a more advanced alternative to traditional parametric spectra for coastal engineering applications.

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